

- Solve $\text{Tr}[M_n D^2 \psi] = \rho_n - \rho$, which is a second-order linear elliptic equation in ψ , on the torus (i.e. with periodic boundary conditions). The solution is unique up to an additive constant (choose for instance the zero-mean solution) provided $M_n > 0$. Call ψ_n the solution.
- Define $\varphi_{n+1} = \varphi_n + \tau \psi_n$ and start again.

From the implementation point of view, the elliptic equation defining ψ , i.e. $\text{Tr}[M_n D^2 \psi] = \rho_n - \rho$ is solved in [218] via a finite-difference discretization of the second order derivatives. It is worthwhile mentioning that this kind of equation, i.e. $\text{Tr}[M D^2 \psi] = f$, where M is the cofactor matrix of an Hessian, can also be expressed in divergence form. Indeed, they read $\sum_{i,j} M^{ij} \psi_{ij} = f$, but $\sum_{i,j} M^{ij} \psi_{ij} = \sum_i (M^{ij} \psi_j)_i$ since $\sum_i M_i^{ij} = 0$ for all j (see Ex(43)). This allows to give a variational structure to this elliptic equation and solve it via finite-element methods. On the other hand, the condition $\rho_n := \det(I - D^2 \varphi_n)$ is more naturally interpreted in terms of second-order finite differences. Let us stress that one of the main difficulties of these equations is exactly the discretization of a non-linear operator such as the Monge-Ampère operator (i.e. the determinant of the Hessian).

The convergence of the algorithm is guaranteed by the following theorem, under regularity assumptions

Theorem 6.4. *Suppose $\rho \in C^{0,\alpha}(\mathbb{T}^d)$ is a strictly positive density and that τ is smaller than a constant c_0 depending on $\|\log \rho\|_{L^\infty}$ and $\|\rho\|_{C^{0,\alpha}}$. Then the above algorithm provides a sequence φ_n converging in C^2 to the solution φ of $\det(I - D^2 \varphi) = \rho$*

Proof. The proof is based on a priori bounds derived from both elliptic regularity theory (see [185], for instance) and Caffarelli's regularity for Monge-Ampère (see Section 1.7.6; for the periodic boundary conditions, see also [128]).

Choosing τ small enough and suitable constants $c_1 > 0$ and C_1, C_2, C_3 large enough, one can prove by induction the following statement: at every step n , we have $I - D^2 \varphi_n > c_1 I$, $\|\log(\rho_n/\rho)\|_{L^\infty} \leq C_1$, $\|\rho_n - \rho\|_{C^{0,\alpha}} \leq C_2$ and $\|\varphi_n\|_{C^{2,\alpha}} \leq C_3$. To do that, note that we have $\|\psi_n\|_{C^{2,\alpha}} \leq C \|\rho - \rho_n\|_{C^{0,\alpha}}$ from standard elliptic regularity theory (see Schauder Estimates in Chapter 6 in [185]). We can compute

$$\begin{aligned} \rho_{n+1} = \det(I - D^2 \varphi_n - \tau D^2 \psi_n) &= \rho_n - \tau \text{Tr}[\text{cof}(I - D^2 \varphi_n) D^2 \psi_n] + r_n \\ &= (1 - \tau) \rho_n + \tau \rho + r_n, \end{aligned}$$

where r_n is the second-order rest in the linearization of the determinant, i.e. it involves second derivatives of φ_n and second derivatives of ψ_n , but is of order 2 in $\tau D^2 \psi_n$. Hence, from the algebra properties of the $C^{0,\alpha}$ norm, we have $\|r_n\|_{C^{0,\alpha}} \leq C \tau^2 \|D^2 \psi_n\|_{C^{0,\alpha}}^2 \leq C \tau^2 \|\rho_n - \rho\|_{C^{0,\alpha}}^2$. Then, from $\rho_{n+1} - \rho = (1 - \tau)(\rho_n - \rho) + r_n$ we get

$$\|\rho_{n+1} - \rho\|_{C^{0,\alpha}} \leq (1 - \tau) \|\rho_n - \rho\|_{C^{0,\alpha}} + C \tau^2 \|\rho_n - \rho\|_{C^{0,\alpha}}^2 \leq \|\rho_n - \rho\|_{C^{0,\alpha}} (1 - \tau + C \tau^2).$$

It is enough to chose τ small enough so that $1 - \tau + C \tau^2 < 1$, to go on with the bound on $\|\rho_{n+1} - \rho\|_{C^{0,\alpha}}$. For the L^∞ bound, we use again

$$\rho_{n+1} = (1 - \tau) \rho_n + \tau \rho + r_n \leq ((1 - \tau)e^{C_1} + \tau + C \tau^2) \rho.$$

Again, for τ small enough and $C_1 > 0$, one can guarantee $(1 - \tau)e^{C_1} + \tau + C\tau^2 \leq e^{C_1}$. The lower bound works in the same way.

In order to study the bound on φ_{n+1} , we write $\varphi_{n+1} = \varphi_n + \tau\psi_n$. Note that, from $I - D^2\varphi_n \geq c_1I$ and $|D^2\psi_n| \leq C$, a suitable choice of τ allows to guarantee the convexity condition $I - D^2\varphi_{n+1} > 0$. Hence, the uniform bound $\|\varphi_{n+1}\|_{C^{2,\alpha}} \leq C_3$ simply follows from Caffarelli's theory, and from the bounds on ρ_{n+1} . And, finally, an upper bound on the Hessian of φ_{n+1} also implies a lower bound on $I - D^2\varphi_{n+1}$ because of the determinant condition.

This shows that the bounds are preserved in the iterations, and the convergence comes from $\|\psi_n\|_{C^{2,\alpha}} \leq C\|\rho - \rho_n\|_{C^{0,\alpha}}$, where the norm $\|\rho - \rho_n\|_{C^{0,\alpha}}$ goes exponentially to 0. $\square$ $\square$

We finish this section by mentioning that, after this first work by Loper and Rapetti, which indeed provided interesting numerical results, many improvement have occurred. First, we underline that [280] extended to the case of a non-constant target density (but still, Lipschitz regularity of the target density was needed). Then, we remark that another strong limitation of [218] is given by its periodic boundary conditions. When working on different domains than $\mathbb{T}^d$ the situation is much trickier. It has been successfully studied by Benamou, Froese and Oberman in [42, 43], by using a monotone discretization of the Monge-Ampère operator. Monotonicity allows to prove the existence of a solution of the discretized problem, and to prove convergence towards the viscosity solution of the limit Monge-Ampère equation. Finally, [42, 43], also attacked the problem of the boundary conditions, and of the shape of the domain. With the approach by viscosity solutions, the authors are able to consider non convex and even non-connected source domains (hence allowing vanishing densities). They handle the second boundary value problem writing the target domain as the level set of a convex function. Yet, we prefer not to enter into details of these recent results, because of their technical difficulty and of the notions about viscosity solutions that we did not develop here.

6.4 Discussion

6.4.1 Discrete numerical methods

We strongly used in Chapter 1 the linear and convex structure of the Kantorovich problem, which is indeed a linear programming problem in infinite dimension. It is natural to look for methods which exploit this fact, after discretization. Here discretization simply means replacing μ and ν with two finitely atomic measures $\mu = \sum_{i=1}^N a_i \delta_{x_i}$ and $\nu = \sum_{j=1}^M b_j \delta_{y_j}$. For fixed N and M , the choice of “optimal” points x_i and y_j and of weights a_i, b_j so that the approximation is as precise as possible is a delicate matter that we do not want to discuss here. It is linked to quantization problems for the measures μ and ν (see [188] for a survey of the theory from a signal processing point of view, and [74] for a variational asymptotical approach). We note anyway that the difficult issue is the choice of the points, since once the points are fixed, the natural choice for the weights is $a_i = \mu(V_i)$, $b_j = \nu(W_j)$, where V and W represent the Voronoi cells of the points $(x_i)_i$ and $(y_j)_j$, respectively (see below).

Box 6.6. – Good to know! – Voronoi cells

Given a finite family of distinct points $(x_i)_i \subset \mathbb{R}^d$, we define their Voronoi cells as follows:

$$V_i = \{x \in \mathbb{R}^d : |x - x_i| \leq |x - x_j| \text{ for all } j\}.$$

These cells are essentially disjoint (only the boundary can be in common: $V_i \cap V_{i'} = \partial V_i \cap \partial V_{i'}$). They are convex polyhedra, possibly unbounded, and the boundary $\partial V_i \cap \partial V_{i'}$ is orthogonal to the vector $x_i - x_{i'}$. Each point x_i obviously belongs to the corresponding V_i .

Given a finite measure ρ which does not give mass to $(d-1)$ -dimensional sets, a Voronoi partition is said to be a *Centroidal Voronoi Tessellation* (CVT) if every point x_i is the barycenter (according to μ) of its cell V_i : $x_i = \frac{1}{\mu(V_i)} \int_{V_i} x d\rho(x)$. Note that, if $\mu = \sum_{i=1}^N a_i \delta_{x_i}$ is a solution of $\min\{W_2(\mu, \rho) : \#(\text{spt}(\mu)) \leq N\}$ (the so-called *location problem*), then the points x_i are a Centroidal Voronoi Tessellation (see **Ex(39)**). There is an efficient algorithm, called *Lloyd Algorithm* which finds a CVT and a local minimier of this optimal quantization problem by iterating a simple construction: “once we have the cells, we move each point to their barycenter” (see [214]).

Anyway, after discretization, the problem becomes finite-dimensional, and the only ingredients are a family of costs c_{ij} (real numbers), and some mass constraints a_i and b_j . The problem becomes

$$\min \left\{ \sum_{i,j} c_{ij} \gamma_{ij} : \gamma_{ij} \geq 0, \sum_i \gamma_{ij} = b_j, \sum_j \gamma_{ij} = a_i \right\}. \quad (6.2)$$

This is a classical linear problem in the standard form $\min\{c \cdot x : x \geq 0, Cx = b\}$ (where $x \geq 0$ means $x_i \geq 0$ for every i). It can be solved for instance with the simplex algorithm.

Box 6.7. – Good to know! – Simplex algorithm

If $c \in \mathbb{R}^d, b \in \mathbb{R}^k, C \in M^{k \times n}(\mathbb{R})$ are given, the minimum in $\min\{c \cdot x : x \in \mathbb{R}^d, x \geq 0, Cx = b\}$, if it exists, is realized by a vertex of the polyhedron $K = \{x \in (\mathbb{R}_+)^d, Cx = b\}$. A vertex is defined as an extremal point of K (i.e. a point x which cannot be written as $x = (y+z)/2$ with $y, z \in K$ and $y \neq z$). In the case of polyhedra, vertices are characterized by the property that the columns C^i of the matrix C corresponding to the indices i with $x_i > 0$ are linearly independent. Every polyhedron has a finite number of vertices.

The simplex algorithm, invented by G. B Dantzig in 1947, considers all possible basis of $\mathbb{R}^k$ (or of the range of C) composed of columns C^i . In this way all vertices are taken into account (one vertex x can correspond to several bases: if the columns C^i with $x_i > 0$ are less than k , then they can be completed at will). The algorithm moves from one basis to another, choosing the next one in a clever way. The choice of the order is based on local improvements of the objective function $c \cdot x$. In this way it can find the optimal vertex (or find that the problem has no solution, if K is unbounded and $c \cdot x$ is not bounded from below on K) without visiting all the vertices.

See [136] and [137] for an interesting history of how linear programming began. Curiously, the example provided by Dantzig to explain how linear programming opened new horizons in optimization is exactly a transport problem (maximal productivity assignment).

Among the good news with this kind of approach there is the fact that, at least in the assignment problem (i.e. the case $N = M$ and $a_i = b_j = 1/N$), one can prove that the vertices of the convex set $\Pi(\mu, \nu)$ are exactly those γ induced by a transport map (which is, in this case, a permutation of $\{1, \dots, N\}$; see Ex(40)).

Among the bad news, there is the computational time of the algorithm. Even if it is much cheaper than visiting all the vertices of $\Pi(\mu, \nu)$, it has in general a huge computational cost⁴.

To overcome this difficulty, one possibility is to exploit the particular form of this linear programming problem, due to the particular features of the matrix C , which only has entries with values 0 and 1. For matrices of this kind there is an interpretation in terms of optimization problems on graphs or *networks*. And in this specific case this is quite easy to understand: the optimal transport from N atoms to M atoms (even when $N \neq M$ and their weights are different) may be seen as a network with $N + M$ nodes (the source and target points) and NM edges (a complete bipartite graph, hence), and the unknown is the flow on each edge, satisfying compatibility with the mass which is imposed at every node. The optimization involves a linear function of these flows. For linear programming problems of this form, an alternative to the simplex algorithm is the so-called *network simplex*, for which we refer to [2] and to its historical reference, [170]. These methods are widely used, even if they also lack worst-case polynomial bounds on their complexity.

Other methods have also been found, also specific to the transport problem and based on duality. Among the most well-known there is the so-called Hungarian Algorithm ([205]), which is polynomial in time, and the Auction algorithm that we describe below.

Auction algorithm This algorithm is also connected with the dual problem, but is not based on a sequence of improvements of the dual objective function, but on a seek for the equilibrium. Due to its connections with the economic considerations of Section 1.7.3, we give a quite detailed description of it (see also [55]).

We specialize to the assignment problem, i.e. the case where the weights a_i and b_j are all equal. We can imagine that we have N buyers, denoted by the index i , and N goods to be bought, denoted by j . We look for an assignment $j = \sigma(i)$, $\sigma \in S_N$ which maximizes⁵ $\sum_i u_{i\sigma(i)}$. The values u_{ij} are the utilities of buyer i when he buys item j . As in Section 1.7.3, we look at a price system $p = (p_j)_j$. Given a price system p and

⁴We sometimes hear that the simplex method is exponential: this is a worst-case estimate. Also, other solvers for linear programming problems exist, for instance interior point methods, and their computational cost can be way better than exponential. However, in practice they are all too expensive.

⁵We switch back to the language, of economists, who prefer to maximize utility rather than minimizing costs.

an assignment σ , we say that it is an equilibrium if for every i we have $u_{i\sigma(i)} - p_i = \max_j u_{ij} - p_j$. The buyers i satisfying this condition are said to be “happy”. This only corresponds to writing the equilibrium condition that we presented in Section 1.7.3 in terms of a coupling γ induced by a permutation. It is well-known that if (p, σ) is an equilibrium, then σ is an optimal assignment (and p is optimal in the dual problem).

Let us start from an arbitrary pair (p_0, σ_0) .

At every step, we have a pair (p^n, σ_n) . If it is an equilibrium then we stop the algorithm. Otherwise, pick any i^* among those i such that $u_{i\sigma(i)} - p_i < \max_j u_{ij} - p_j$. The selected buyer i^* implements two actions:

1. he chooses one good j^* realizing the maximum in $\max_j u_{i^*j} - p_j$ and exchanges his own good $\sigma_n(i)$ with the buyer $\sigma_n^{-1}(j^*)$ who was originally assigned to j^*
2. he increases the price of j^* to a new value which is such that he is indifferent between j^* and the second best object in $\max_j u_{i^*j} - p_j$.

Thus, the new price of j^* is $P = p_{j^*}^n + \max_j (u_{i^*j} - p_j^n) - \max_{j \neq j^*} (u_{i^*j} - p_j^n) \geq p_{j^*}^n$.

The new price system p^{n+1} is defined by setting $p_j^{n+1} = p_j^n$ for $j \neq j^*$ and $p_{j^*}^{n+1} = P$. The new permutation σ_{n+1} is obtained by composing with the transposition which exchanges j^* and $\sigma_n(i^*)$.

Unfortunately, the above algorithm can loop forever, because of possible ex-aequo in $\max_j u_{i^*j} - p_j$. Indeed, in such a case the prices can stay constant, and a certain subset of buyers could go on exchanging among them the same set of goods. To bypass this problem, it is useful to introduce a tolerance parameter $\varepsilon > 0$. We define a buyer i to be ε -happy whenever $u_{i\sigma(i)} - p_i > \max_j (u_{ij} - p_j) - \varepsilon$. We look now for a pair (p, σ) where all buyers are ε -happy. The algorithm evolves as before, selecting a buyer i^* who is not ε -happy, letting him exchange a good with $\sigma_n^{-1}(j^*)$, but the rule for price increase changes. Indeed, we can now set $P = p_{j^*}^n + \max_j (u_{i^*j} - p_j^n) - \max_{j \neq j^*} (u_{i^*j} - p_j^n) \geq p_{j^*}^n + \varepsilon$. This price is the maximal one that we can give to item j^* so that buyer i^* is ε -happy if he purchases j^* at such a price. The advantage is that we have now $P \geq p^n(j^*) + \varepsilon$ and there is a minimal increment.

It is easy to see an analogy with auction systems, and that this ε -system imposes a minimal bid.

Now, we can see that every time that an item is chosen by a buyer and receives a bid, then, from that moment on, whoever owns such an item will be ε -happy. Indeed, at the very moment when a buyer gets an object by a bid, he is ε -happy, and the price of such an object cannot increase as long as he owns it, as well as the prices of the others cannot decrease. Hence, the algorithm will stop as soon as all items will have received at least a bid each. Now, suppose that there is an item which has received no bids: his price is constant, while the prices of the other increase. It is not difficult to prove that the number of possible iterations going on in the presence of an object which has never received a bid must be bounded (as soon as there is one unbidded good, all the others become too expensive after a certain number of received bids, and when they become too expensive they receive no more bids). This means that the algorithm stops after a finite number of steps.

Once we have a pair (p, σ) which is a equilibrium in the sense that every buyer is ε -happy, then we can prove that the cost $\sum_i u_{i\sigma(i)}$ is almost optimal among permutations

$\tilde{\sigma}$. Indeed, take any other permutation $\tilde{\sigma}$. We have

$$u_{i\sigma(i)} - p_i \geq \max_j (u_{ij} - p_j) - \varepsilon \geq u_{i\tilde{\sigma}(i)} - p_{\tilde{\sigma}(i)} - \varepsilon.$$

Summing over i , and using $\sum_i p_i = \sum_i p_{\tilde{\sigma}(i)}$, we get

$$\sum_i u_{i\sigma(i)} \geq \sum_i u_{i\tilde{\sigma}(i)} - N\varepsilon.$$

This proves that σ is optimal, up to an error of $N\varepsilon$. By the way, if the utilities u_{ij} are integer numbers and $N\varepsilon < 1$, then σ must actually be optimal, with no error.

A rough estimate of the number of iterations which are necessary to conclude the algorithm is obtained in the following way: set $C = \max |u_{ij}|$ and start with $p^0 = 0$. Till there is at least an unbidded good, no other good can receive more than C/ε bids. This means that the number of iterations is at most of the order of NC/ε . In the case of integer utilities, an exact solution is found in $O(CN^2)$ iterations⁶.

We remark that this cost can be indeed very high because of the presence of the factor C . Indeed, if the costs are rational numbers between 0 and 1 with k digits, if we want to make them integers we could be lead to multiply up to $C = 10^k$. A way to circumvent this problem is to make an iterative ε -scaling procedure. We first find an assignement which is $N\varepsilon$ -optimal for a certain value of ε , then we divide ε by two, and we go on. This procedure, introduced in [57], is also detailed in [234]. We refer to these notes, and to [56] for considerations about possible accelerated variants and computational costs of this method, very efficient in many cases. We also mention that in Section 8.4.4 we will briefly describe a cosmological model which has been numerically attacked thanks to the auction algorithm.

Entropic regularization and iterated projections We would like to discuss another, very recent, but simple to explain, approach to the discrete linear programming version of the optimal transport problem. The starting point is the following variant of the linear problem (6.2): fix $\varepsilon > 0$ and look at

$$\min \left\{ \sum_{i,j} (c_{ij}\gamma_j + \varepsilon\gamma_j \log(\gamma_j)) : \gamma_j \geq 0, \sum_i \gamma_j = b_j, \sum_j \gamma_j = a_i \right\}. \quad (6.3)$$

It is clear that, for $\varepsilon \rightarrow 0$, the above minimization problem converges (in the sense of Γ -convergence, see Box 4.6) to the minimization in (6.2). We try to describe now why this approximated problem is easier to handle.

First, we rewrite the objective function using

$$c_{ij}\gamma_j + \varepsilon\gamma_j \log(\gamma_j) = \varepsilon\gamma_j \log\left(\frac{\gamma_j}{\eta_{ij}}\right) \quad \sum_{i,j} (c_{ij}\gamma_j + \varepsilon\gamma_j \log(\gamma_j)) = \varepsilon \text{KL}(\gamma|\eta),$$

⁶Observe that this assumption on the values u_{ij} being integer, corresponding to integer costs on each edge in the network interpretation, appears quite often and simplifies the complexity of other algorithms as well. For instance, in [245] a polynomial bound on the cost of a network simplex algorithm is given under this assumption, obtaining $O(\min(kh \log(kC), kh^2 \log k))$ for a network with k nodes, h edges, and maximal cost equal to C .

where $\eta_{ij} = e^{-c_{ij}/\varepsilon}$ and KL denotes the so-called *Kullback-Leibler* divergence. This is not at all a divergence in the sense of differential calculus, but rather a sort of distance based on the relative entropy:

$$\text{KL}(\gamma|\eta) := \sum_{i,j} f\left(\frac{\gamma_{ij}}{\eta_{ij}}\right) \eta_{ij} \quad \text{for } f(t) = \begin{cases} t \log t & \text{if } t \geq 0, \\ +\infty & \text{if } t < 0. \end{cases}$$

Hence, the above minimization problem reads as the projection, for the Kullback-Leibler divergence, of the point $\eta \in \mathbb{R}^{N \times N}$ on the set of constraints $C = C^x \cap C^y$, where $C^x = \{\gamma \in \mathbb{R}^{N \times N} : \sum_j \gamma_{ij} = a_i\}$ and $C^y = \{\gamma \in \mathbb{R}^{N \times N} : \sum_i \gamma_{ij} = b_j\}$. It is important to notice that the positivity constraint on γ has disappeared, because it is included in the definition of the entropy function f , which acts as a sort of barrier for this constraint (it is not a true barrier in the usual sense because it does not tend to infinity as the variable tends to 0; yet, it is set to infinity on negative numbers, and the derivative tends to infinity at 0).

The Kullback-Leibler divergence is not a distance, but shares some of the properties of distances. In particular it can be proven that the projection onto the intersection of two linear subspaces such as C^x and C^y can be obtained by alternate projections, i.e. defining $\gamma^{2k+1} = \text{Proj}_{C^x}(\gamma^{2k})$ and $\gamma^{2k+2} = \text{Proj}_{C^y}(\gamma^{2k+1})$, with $\gamma^0 = \eta$, and looking at the limit as $k \rightarrow \infty$.

The problem becomes that of the projection on one on the two subspaces, but this happens to be doable explicitly. Given a certain $\bar{\gamma}$, we want to solve (for instance), the minimization problem

$$\min \left\{ \sum_{i,j} \gamma_{ij} \log \left(\frac{\gamma_{ij}}{\bar{\gamma}_{ij}} \right) : \sum_j \gamma_{ij} = a_i \right\}.$$

It can be seen that the problem is separable according to the variable i (i.e. that every i can be treated separately, and the optimality condition gives $\log(\gamma_{ij}) = \log(\bar{\gamma}_{ij}) + \lambda_i$, i.e. $\gamma_{ij} = p_i \bar{\gamma}_{ij}$, for some constants p_i which are determined thanks to the constraint (i.e. $p_i = a_i / (\sum_j \bar{\gamma}_{ij})$).

Analogously, it can be proven that the projection onto C^y has the form $\gamma_{ij} = q_j \bar{\gamma}_{ij}$, for $q_j = b_j / (\sum_i \bar{\gamma}_{ij})$.

As a consequence, one can solve the approximated problem by means of a sequence of iterations of the following algorithm: start from $\gamma^0 = \eta$, define $\gamma_{ij}^{2k+1} = p_i \gamma_{ij}^{2k}$ with $p_i = a_i / (\sum_j \gamma_{ij}^{2k})$, and then $\gamma_{ij}^{2k+2} = q_j \gamma_{ij}^{2k+1}$ with $q_j = b_j / (\sum_i \gamma_{ij}^{2k+1})$. The limit as $k \rightarrow \infty$ provides the desired solution of the “regularized” problem (the regularization being given by the addition of the entropy).

This idea, called *iterative proportional fitting procedure* (IPFP), can be traced back to [143] and even [282], and it has been recently applied to many optimal transport problems in [40]. It shows that the choice of the entropic regularization has several advantages:

- it allows to get rid of the positivity constraint (which would not be the case for a quadratic regularization);



Figure 6.2: The optimal γ in a 1D transport problem (the starting measure is a centered Gaussian, the arrival measure is composed of two bumps), via IPFP methods with decreasing values of ε . The image on the right gives a quite precise description of the optimal (monotone) map. Pictures kindly provided by L. Nenna.

- it allows to compute explicitly the projection at every step, as soon as we only project on one subspace C^x or C^y only;
- it is not too costly in terms of memory storage, since at every step the current γ^{2k} is of the form $\gamma_{ij}^{2k} = \eta_{ij} p_i q_j$ (cumulating the product of the coefficients p_i and q_j appearing at every step), which allows to stock $2N$ data instead of N^2 ;
- it is not too costly at every step, as it only requires $O(N)$ operations;
- it can be easily parallelized, due to the separable structure of each step;
- it can be adapted to multi-marginal problems and other linear problems.

It is also possible to deduce, from the limit value of the coefficients p and q , the Kantorovich potentials ϕ and ψ , but we leave it as an exercise (see **Ex(42)**).

6.4.2 Semidiscrete numerical methods

After the study of two discrete measures, the case of one discrete measure and one continuous deserve a short discussion. We are speaking of the case where $\mu = f(x) dx$ has a density and $\nu = \sum_{j=1}^N b_j \delta_{y_j}$ is finitely atomic.

The main ideas behind the methods that are used rely on the dual problem, in the following formulation

$$\max \left\{ F(\psi) = \sum_j \psi_j b_j + \int_{\Omega} \psi^c(x) f(x) dx \right\},$$

where the numbers ψ_j represent the values of the function ψ at the points y_j and ψ^c is the c -transform of ψ . The considerations in Chapter 1 explain why solving this problem gives information on the primal problem $\min\{\int c d\gamma : \gamma \in \Pi(\mu, \nu)\}$. In particular, once we find the optimal function ψ , the set of points x satisfying $\psi_j + \psi^c(x) = c(x, y_j)$ will

provide the set of points transported onto the target point y_j . We note that $\psi^c(x) = c(x, y_j) - \psi_j$ exactly means that $c(x, y_j) - \psi_j \leq c(x, y_{j'}) - \psi_{j'}$ for all j' . This gives rise to the study of some sorts of Voronoi cells, whose definition also involves some scalar coefficients ψ_j . We will analyze these cells in the simpler case where $c(x, y) = \frac{1}{2}|x - y|^2$. In this case we define

$$V_\psi(j) = \left\{ x : \frac{1}{2}|x - y_j|^2 - \psi_j \leq \frac{1}{2}|x - y_{j'}|^2 - \psi_{j'} \text{ for all } j' \right\}.$$

By expanding the squares we see that each $V_\psi(j)$ is defined by $N - 1$ linear inequalities of the form $x \cdot (y_{j'} - y_j) \leq a(j, j') := \psi_j - \psi_{j'} + \frac{1}{2}|y_{j'}|^2 - \frac{1}{2}|y_j|^2$ and is hence a convex polyhedron. These cells are called *power cells*. They coincide with the Voronoi cells of the points y_j whenever all the values ψ_j are equal.

The problem consists in finding values ψ_j such that the masses $\int_{V_\psi(j)} f(x) dx$ equal the prescribed values b_j . This is indeed equivalent to maximizing the quantity $F(\psi)$. The maximization of F is a finite-dimensional issue, since (thanks to v being finitely atomic) the only values of ψ which are relevant are those taken at the points y_j . Hence, we can consider $\psi \in \mathbb{R}^N$, and look at the derivatives of F .

First, we note that F is concave in ψ . Indeed, the first part of F is linear: $\psi \mapsto \sum_j \psi_j b_j$. Then, we take the values $\psi^c(x) = \inf_j c(x, y_j) - \psi_j$; these values are infima of linear functions of ψ , and thus are concave. They are weighted with a positive density f and then integrated, the result is a concave function of ψ . Let us compute the derivative of F with respect to one variable ψ_j : from

$$\frac{\partial \psi^c(x)}{\partial \psi_j} = \begin{cases} -1 & \text{if } c(x, y_j) - \psi_j < c(x, y_{j'}) - \psi_{j'} \text{ for all } j' \neq j, \\ 0 & \text{if } c(x, y_j) - \psi_j > \psi^c(x), \\ \text{not defined} & \text{if } c(x, y_j) - \psi_j = \psi^c(x) = c(x, y_{j'}) - \psi_{j'} \text{ for some } j' \neq j, \end{cases}$$

and from the fact that the last of these three cases is negligible (at least in the case $c(x, y) = |x - y|^2$ where the equality case reduces to a subset of an hyperplane), we deduce that F is indeed differentiable everywhere, and⁷

$$\frac{\partial F}{\partial \psi_j} = b_j - \int_{V_\psi(j)} f(x) dx.$$

This shows that the maximization of F is equivalent to $b_j = \int_{V_\psi(j)} f(x) dx$, and also allows to look for the maximum of F via gradient methods. To this aim, it is also useful to check upper bounds on the second derivatives of F . We restrict again our attention to the case $c(x, y) = \frac{1}{2}|x - y|^2$. We note that moving the value of the coordinates of ψ from ψ_j to $\psi_j + h_j$ makes all the cells $V_\psi(j')$ change, but only by changing the constants $a(j, j')$ of a value $h_{j'} - h_j$. This means that the hyperplanes defining their boundaries move of a distance of the order of $\frac{|h|}{|y_j - y_{j'}|}$. If the points y_j are fixed and distinct, the

⁷The computation of the gradient, and its continuity, could have been simplified by looking at the sub-differential, see [25]

measure of the symmetric difference $V_\psi(j')\Delta W_{\psi+h}(j')$ is bounded by $C|h|$. If f is L^∞ , then one obtains

$$\left| \frac{\partial F}{\partial \psi_j}(\psi) - \frac{\partial F}{\partial \psi_j}(\psi + h) \right| \leq C|h|, \quad (6.4)$$

and $F \in C^{1,1}$.

Box 6.8. – Memo – Gradient algorithm

If $F : \mathbb{R}^N \rightarrow \mathbb{R}$ is a differentiable convex function, an easy algorithm to look for the minimum of F is the following: take $x_0 \in \mathbb{R}^N$ and $\tau > 0$ a fixed size, and define

$$x_{k+1} = x_k - \tau \nabla F(x_k).$$

If F is such that $\alpha I \leq D^2 F(x) \leq LI$ for every x and some $\alpha, L > 0$, then the above algorithm converges exponentially whenever $\tau < 2\alpha/L^2$. More precisely, F is strictly convex and has a unique minimizer $\bar{x}$, and we have $|x_k - \bar{x}| \leq C\beta^k$, where β depends on τ, α, L and $\beta < 1$ if $\tau \in]0, 2\alpha/L^2[$. The convergence $F(x_k) - \min F$ is also exponential.

If $D^2 F$ is bounded from above but not from below (i.e. F is smooth but not enough convex), but admits a minimizer $\bar{x}$, then the rate of convergence is much worse: indeed, one can only prove $F(x_k) - F(\bar{x}) \leq C/k$, as soon as $\tau M < 1$.

Many texts present a wide introduction to gradient methods in convex optimization, see for instance [65] for more details. Variants where the step size τ varies with k , both in prescribed and optimized manners, also exist.

As a final point, note that the above algorithm also corresponds to an explicit Euler scheme in the discretization of the gradient flow of F (see the Memo Box 8.1).

Then, the practical implementation of a gradient algorithm of this kind depends on how one can practically compute the power cells $V_\psi(j)$ and the integral of the density f on them. This is a matter of the so-called *computational geometry*. We do not spend word on this very delicate issue, but refer for instance to the recent papers [233, 210] for a detailed discussion. The original implementation of these methods is proposed in [233], and the implementation has been improved in [210], where and spectacular numerical simulations⁸ of optimal transport between geometric shapes are also presented. Some pictures are presented here below in Figures 6.4.2 and 6.4.2.

Due to the very good numerical results, that we present through figures 6.4.2 and 6.4.2, we prefer to spend some extra words on the most efficient methods to be used in this semi-discrete setting.

Indeed, even if the description of the algorithm was easier in the case of a gradient algorithm, the convergence is much quicker if one uses instead a Newton approach. Newton's method (see Box 6.4 in Section 6.3) is suitable to find zeros of a vector function from $\mathbb{R}^N$ to $\mathbb{R}^N$, which is in this case the function associating to a dual potential $\psi = (\psi_i)_i \in \mathbb{R}^N$ the values $b_i - \int_{V_\psi(i)} f(x) dx = \partial F / \partial \psi_i$. In order to be able to use the Newton method, one needs to be able to compute their derivatives, which are indeed the entries of the Hessian matrix of F . One also needs to justify that this Hessian

⁸The pictures in the next pages have been kindly provided by B. Lévy and are produced with the same algorithms as in [210].

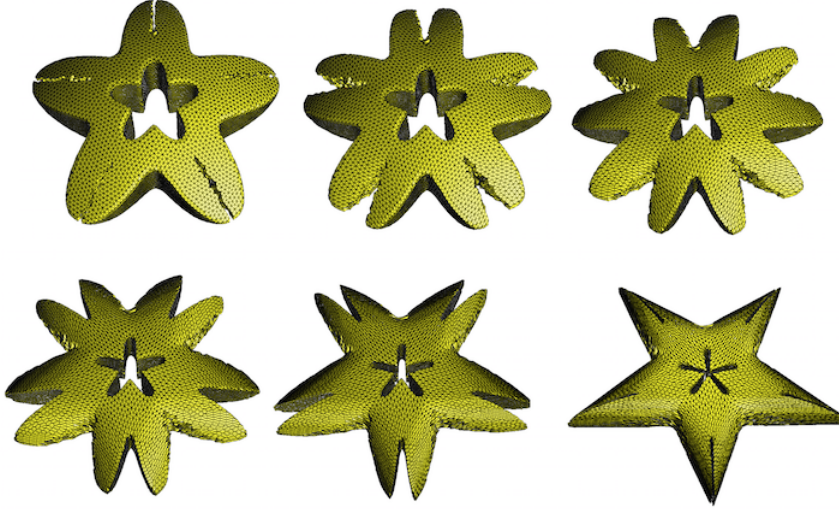


Figure 6.3: W_2 geodesic between two different stars in 3D, computed via semi-discrete optimal transport on a simplex triangulation. Once obtained the images of every vertex x_i of the triangulation, linear interpolation between each x_i and $T(x_i)$ is computed. Note the topology change and the singularities of the optimal map on every branch of the star.

matrix is non-singular, and provide lower bounds on its eigenvalues. This corresponds to uniform convexity of F .

Moreover, we know that the Newton method is only efficient when we are able to choose a suitable initialization for the algorithm, since it only converges starting from a certain neighborhood. This is efficiently addressed in [233], where the problem is considered with a multi-resolution point of view. We associate to the family of points y_j a number of smaller families, each obtained from the previous one via a Lloyd algorithm and dividing the number of points by a fixed ratio. In this way there is a correspondence such that every point in a family has a unique image in the next one (which is smaller). We start from the computation of a potential ψ for the smallest family, which is easy to do. Then, at every step, the potential ψ computed for the i -th family is used to build an initialization for the Newton method of the next step, the one for the $(i - 1)$ -th family of points. This guarantees that we are always sufficiently close to the minimizer that we look for, and that the algorithm converges in a very small number of iterations. In practice, the implementations of [210] can handle one million points in some minutes.

We come back now to the justification of uniform convexity, which is required to apply the Newton procedure. Actually, the function F is far from being strictly convex: indeed, it does not change if one adds a same constant to all the ψ_i . Yet, up to this invariance, we can expect it to be strictly convex. In order to do that, one needs to fix a value $\psi_{i_0} = 0$, and to look at the behavior w.r.t. ψ_i for $i \neq i_0$.

How to compute the second derivatives of F ? by improving the computations lead-

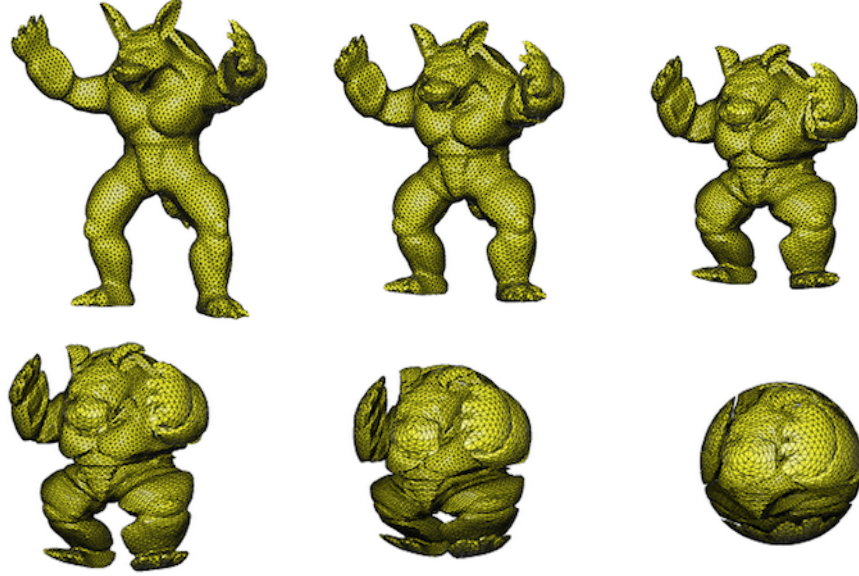


Figure 6.4: W_2 geodesic between the uniform measure on the shapes of an “armadillo” and of a sphere (discretized with up to a million of Dirac masses).

ing to (6.4), if f is a continuous density, then one obtains

$$\frac{\partial^2 F}{\partial \psi_j \partial \psi_j}(\psi) = -\frac{1}{|y_i - y_j|} \int_{\Gamma_{ij}} f(x) d\mathcal{H}^{d-1}(x), \quad \text{for } i \neq j,$$

where $\Gamma_{ij} = \partial V_\psi(i) \cap \partial V_\psi(j)$ is the common boundary (if any) of the two cells corresponding to the indices i and j (which is $(d-1)$ -dimensional polyhedron). For the pure derivatives, one has

$$\frac{\partial^2 F}{\partial^2 \psi_i}(\psi) = \sum_{j \neq i} \frac{1}{|y_i - y_j|} \int_{\Gamma_{ij}} f(x) d\mathcal{H}^{d-1}(x) = -\sum_{j \neq i} \frac{\partial^2 F}{\partial \psi_j \partial \psi_j}(\psi).$$

Note that in these sums we also keep the index i_0 . This allows to write a Hessian in N variables which is singular since the sum of every line is 0. Then, we withdraw the line and column corresponding to the index i_0 , and we have a square $(N-1) \times (N-1)$ matrix M . In order to check invertibility of M , we use the following lemma.

Lemma 6.5. *Any matrix $M \in \mathbb{M}^{k \times k}$ satisfying the following properties is invertible.*

(H1) *for all i , we have $M^{i,i} \geq \sum_{j \neq i} |M^{i,j}|$,*

(H2) *there exists i such that $M^{i,i} > \sum_{j \neq i} |M^{i,j}|$,*

(H3) *for any pair (i, j) there is a sequence $i_1, i_2, \dots, i_k$ with $i_1 = i$, $i_k = j$ satisfying $M^{i_h, i_{h+1}} \neq 0$.*

Proof. Let $x \in \text{Ker}(M)$ and let i_0 be an index such that $|x_i|$ is maximal. We may suppose for simplicity that x_{i_0} is positive. Then we have

$$0 = M^{i_0, i_0} x_{i_0} - \sum_{j \neq i_0} M^{i_0, j} x_j \geq M^{i_0, i_0} x_{i_0} - \sum_{j \neq i_0} |M^{i_0, j}| x_{i_0} = x_{i_0} \left(M^{i_0, i_0} - \sum_{j \neq i_0} |M^{i_0, j}| \right) \geq 0.$$

This implies that all inequalities are equalities and in particular $x_j = x_{i_0}$ whenever $M^{i_0, j} \neq 0$. Hence, the entries of x on all the indices which are “neighbors” of i_0 equal x_{i_0} (and they are maximal as well). This allows to repeat the argument replacing i_0 with another maximizing index j and so on... since any index is connected by a chain of neighbors to i_0 , we get that all the entries are equal. But this implies that the vector that we selected in the kernel must be a multiple of the vector $(1, 1, \dots, 1)$. Yet, this vector is not in the kernel since the sum of the elements on each line is not zero for all lines, by assumption (H2). This proves that M is invertible. $\square$ $\square$

The above lemma can be applied to the Hessian of F under some reasonable assumptions (for instance, we should require that the support of the density f is arcwise connected).

This guarantees that the Hessian of F is a strictly positive definite matrix (on the space $\mathbb{R}^{N-1}$ of potentials with fixed $\psi_{i_0} = 0$) at every point. In order to apply a Newton’s method one should also bound from below its eigenvalues. This is more delicate, and we refer to [114] both for a counterexample under the three assumptions of the above Lemma, which are not enough for such a bound, and for a proof in the specific case where the matrix M comes from a partition into power cells.

Moreover, as we are dealing with the arguments of [114], let us underline another semi-discrete method, described in the same paper. The idea of the method comes from the considerations of our Section 2.4 about Knothe transport. Indeed, as we proved that the Knothe map is the limit of the optimal transport maps for degenerate quadratic costs corresponding to a matrix $\text{diag}(1, \varepsilon, \varepsilon^2, \dots, \varepsilon^{d-1})$, we can start from such a map (Knothe) and then let ε vary. The Knothe map is easy to compute, and one can find the cells corresponding to it⁹. Since for the Knothe map the most important coordinate is the last one, these cells are horizontal strips with prescribed volume. Note that we need to suppose that they have different x_d components for the method to work. After the computation of the initial condition, one runs an ODE in dimension $N - 1$ (N being the number of atoms at the arrival), and finds the evolution of the Kantorovich potential $\psi(\varepsilon)$. The computation to prove that the ODE is well-posed are similar to the ones above and allow to conclude that the evolution fits into the standard existence, uniqueness, stability and approximation theory for Cauchy problems with Lipschitz

⁹This is the semidiscrete method coming from this idea, implemented in [114]. There is also a continuous counterpart, i.e. a PDE on the Kantorovich potentials ψ_ε . It has been studied from the theoretical point of view in [67], where Bonnotte proved a non-trivial well-posedness result, based on the application of the Nash-Moser implicit function theorem around $\varepsilon = 0$ (once we are on $\varepsilon > 0$ the equation is more regular). Later, the same author also started numerical implementations of the time-discretization of this equation in [66]. Note that his continuous method recalls in some aspects both the AHT flow described in Section 6.2 (as it starts from the Knothe map, is continuous in time but discretized for numerical purposes, and evolves by imposing prescribed image measure at every time) and the LR Newton’s iterations of Section 6.3 (as the underlying PDE is based on a linearization of the Monge-Ampère equation).

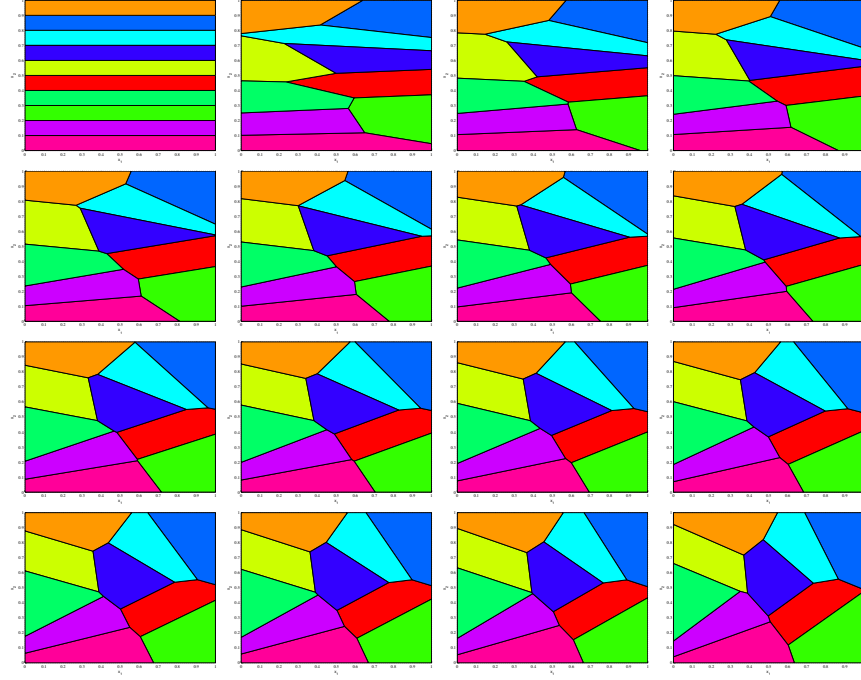


Figure 6.5: Ten sample points: evolution of the tessellation for $\varepsilon = 0$ to $\varepsilon = 1$ (from top left to bottom right). Picture taken from the preliminary studies leading to [114].

vector fields. Hence, one can apply a time-discretization in the form of an explicit Euler scheme to approximate the solution. Observing the evolution of the potentials ψ_ε as ε runs from 0 to 1 we can also compute the corresponding cells, which move from rectangular strips to convex polyhedral power cells.

We show in Figure 6.5 a simple sequence of cells obtained with the above method.

Chapter 7

Functionals on the space of probabilities

We consider in this chapter various classes of functionals on the space $\mathcal{P}(\Omega)$, which can be of interest in many variational problems, and are natural in many modeling issues. Indeed, in several applied models, we can face minimization problems where the unknown to be determined is the distribution of a certain amount of mass, and the criteria involve one or more of the following functionals.

- The integral of a given function (*potential energy*)

$$\mathcal{V}(\mu) := \int_{\Omega} V \, d\mu.$$

- The double integral of a function on $\Omega \times \Omega$ according to the tensor product $\mu \otimes \mu$ (*interaction energy*)

$$\mathcal{W}(\mu) := \int_{\Omega \times \Omega} W(x, y) \, d\mu(x) \, d\mu(y).$$

- The Wasserstein distance W_p (or a function of it) from a fixed measure ν ; for simplicity, we consider rather the p -th power of W_p :

$$\mu \mapsto W_p^p(\mu, \nu).$$

More generally, we will also consider transport cost $\mathcal{T}_c(\mu, \nu)$ defined as usual as $\min\{\int c(x, y) \, d\gamma, \gamma \in \Pi(\mu, \nu)\}$, for general costs $c(x, y)$.

- The norm in a dual functional space: given a Banach space of functions on Ω , we take

$$\|\mu\|_{\mathcal{X}'} := \sup_{\varphi \in \mathcal{X}, \|\varphi\| \leq 1} \int \varphi \, d\mu = \sup_{\varphi \in \mathcal{X} \setminus \{0\}} \frac{\int \varphi \, d\mu}{\|\varphi\|}.$$

- The integral of a function of the density

$$\mathcal{F}(\mu) := \begin{cases} \int_{\Omega} f(\rho(x)) dx & \text{if } \mu = \rho \cdot dx, \\ +\infty & \text{otherwise.} \end{cases}$$

- The sum of a function of the masses of the atoms

$$\mathcal{G}(\mu) := \begin{cases} \sum_i g(a_i) & \text{if } \mu = \sum_i a_i \delta_{x_i}, \\ +\infty & \text{if } \mu \text{ is not purely atomic.} \end{cases}$$

The scope of this chapter is to study some properties of these functionals. The first questions that we analyze are classical variational issues (semi-continuity, convexity, first variation...). Then, we also introduce and analyze a new notion, the notion of geodesic convexity. This is a natural concept in the analysis on metric spaces: is a certain functional convex *along geodesics* of the metric space? In the case of the Wasserstein spaces, and in particular of $\mathbb{W}_2$, this concept, introduced by McCann in [229], is called *displacement convexity*. It turns out to be quite different than usual convexity, but very useful in many cases.

7.1 semi-continuity

Since the main goal is to use these functionals in variational models, in order to be able to minimize them, we first give some semi-continuity criteria. We will consider semi-continuity or continuity of these functionals with respect to weak convergence (meaning as usual the duality with C_b functions). We start from the most simple functionals.

7.1.1 Potential & interaction energies, transport costs, dual norms

We start this review from the easiest case: the potential energy.

Proposition 7.1. *If $V \in C_b(\Omega)$ then $\mathcal{V}$ is continuous for the weak convergence of probability measures. If V is l.s.c. and bounded from below then $\mathcal{V}$ is semi-continuous. Moreover, semi-continuity of V (respectively, continuity) is necessary for the semi-continuity (continuity) of $\mathcal{V}$.*

Proof. The continuity of $\mu \mapsto \int V d\mu$ for $V \in C_b$ is straightforward by definition of weak convergence. If V is l.s.c. and bounded from below, we know that there is a sequence of Lipschitz and bounded functions V_k increasingly converging to V (see Box 1.5). Then, by monotone convergence, we infer that $\mathcal{V}(\mu) = \lim_k \int V_k d\mu = \sup_k \int V_k d\mu$. This allows to see $\mathcal{V}$ as a supremum of continuous functionals, and hence it is l.s.c.

The necessity part is straightforward if one considers sequences of points $x_k \rightarrow x$ and the associated Dirac masses. Since $\delta_{x_k} \rightarrow \delta_x$ then continuity of $\mathcal{V}$ implies $V(x_k) \rightarrow V(x)$ and semi-continuity implies $\liminf_k V(x_k) \geq V(x)$. $\square$ $\square$